Software Design Diagram Generator

Product Backlog

IT314 Software Engineering · Group 27 · Draft

1. Summary

ID	User story	Epic
US1	User Registration	Epic 1 — Account
US2	User Login	Epic 1 — Account
US3	Create Project	Epic 2 — Projects
US4	View and Open Projects	Epic 2 — Projects
US5	Provide Requirements	Epic 3 — Requirements
US6	Extract Requirements Information	Epic 4 — Requirement Extraction
US7	Review Extracted Information	Epic 5 — Review & Confirmation
US8	Edit Extracted Information	Epic 5 — Review & Confirmation
US9	Confirm Requirements Model	Epic 5 — Review & Confirmation
US10	Generate Use Case Diagram	Epic 6 — Diagram Generation

Sprint 1 flow: Register/Login (US1–US2) → Create/Open Project (US3–US4) → Provide Requirements (US5) → Extract (US6) → Review (US7) → Edit (US8) → Confirm (US9) → Generate Use Case Diagram (US10). Each stage depends on the one before it.

2. User stories

Epic 1 — Account

US1 — User Registration	
As a user, I want to create an account so that I can access the application and save my projects.	
Success	**Failure**
• Required details are validated; password has at least 8 characters with a letter and a digit. • Account is created and a verification email is sent (link valid 24 hours). • Login is possible only after verification.	• Missing or invalid details are rejected with a clear error. • A duplicate email or expired link shows a clear message.
Tasks: FR28a, TE5, X14, X9	

US2 — User Login	
As a user, I want to log in securely so that I can access my projects and continue my work.	
Success	**Failure**
• Valid credentials lead to the project area. • Session expires after 24 hours of inactivity. • Logout ends the session on the server at once.	• Incorrect credentials are rejected without revealing which field was wrong. • Five failed attempts lock the account for 15 minutes. • Unauthenticated access redirects to login; API returns 401.
Tasks: FR28c, FR28d, NFR4a, X14	

Epic 2 — Projects

US3 — Create Project	
As a user, I want to create a project so that I can organize my requirements and generated diagrams.	
Success	**Failure**
• Name of 1–100 characters (trimmed) creates a project owned by the user. • The new project can be opened.	• A missing or invalid name is rejected with what to correct. • A duplicate name (case-insensitive) for the same owner is rejected.
Tasks: FR3a	

US4 — View and Open Projects	
As a user, I want to view and open my projects so that I can continue working on an existing project.	
Success	**Failure**
• Owned projects are listed, newest first. • With none, a "Create your first project" prompt is shown. • The selected project opens.	• A project that is missing or inaccessible returns "not found". • Incorrect project data is never opened.
Tasks: FR3c, X6	

Epic 3 — Requirements

US5 — Provide Requirements	
As a user, I want to provide my software requirements so that the application can analyse them and prepare them for modelling.	
Success	**Failure**
• The user types or pastes requirement text. • Consent is recorded (timestamp, policy version) before the first submission. • Valid text is split into numbered statements (R-001, …) and saved to the project.	• Empty or whitespace-only text is rejected and nothing is saved. • Without consent, nothing is sent to the AI provider.
Tasks: FR1a, NFR5a, NFR5d, X8	

Epic 4 — Requirement Extraction

US6 — Extract Requirements Information	
As a user, I want the application to extract relevant information from my requirements so that I do not have to identify the modelling information manually.	
Success	**Failure**
• Six categories are extracted: actors, entities, processes, data flows/stores, interactions, relationships (median target 60 s). • Each item links to its source statement(s). • Uncertain items are kept and flagged, not dropped.	• A failed extraction is reported; an incomplete result is not shown as a success. • An empty category shows an explanatory note.
Tasks: TE4, X10, FR4g, FR4a–f, X11, S3	

Epic 5 — Review & Confirmation

US7 — Review Extracted Information	
As a user, I want to review the extracted information so that I can verify that it correctly represents my requirements.	
Success	**Failure**
• All categories are shown grouped in one place; over 200 items paginate at 50. • The user can identify items needing correction.	• If review data is unavailable, the user is told and confirmation is blocked.
Tasks: FR5a	

US8 — Edit Extracted Information	
As a user, I want to edit incorrect or incomplete extracted information so that the final model accurately represents my requirements.	
Success	**Failure**
• The user can modify, add and remove items. • Edits save immediately and are marked “user-edited”.	• Invalid, incomplete or empty edits are rejected with a prompt to correct.
Tasks: FR5b	

US9 — Confirm Requirements Model	
As a user, I want to confirm the reviewed requirements model so that the approved information can be used to generate a diagram.	
Success	**Failure**
• The reviewed model can be confirmed and becomes the input for generation. • Generate is disabled until confirmed; any later change clears the confirmation.	• An incomplete review blocks confirmation and shows what to correct. • Skipping confirmation by URL or API call is rejected by the server.
Tasks: FR5e	

Epic 6 — Diagram Generation

US10 — Generate Use Case Diagram	
As a user, I want to generate a Use Case diagram from my confirmed model so that I can see and export my system's actors and use cases.	
Success	**Failure**
• The user selects Use Case; the selection is locked for the run, with progress shown throughout. • Actors and use cases match the extraction; include/extend are dashed arrows with stereotypes. • Elements with no linked requirement are tagged "assumption". • The diagram is saved to the project and viewable; clicking an element shows its sources. • PNG exports at 2× scale, capped at 8,000 px.	• An unconfirmed model blocks generation. • A failed run is reported; no incomplete diagram is shown as a success. • Insufficient data gives a partial diagram with assumption markers.
Tasks: FR9d, X7, FR16a, FR10a, DR1a, DR2a, X12, FR18a, FR10g, X15, S1	

3. Later sprints

Item	**Sprint**
PDF and DOCX upload	2
Flags and validation	2
Class and Activity diagrams	2
Google sign-in (FR28b)	2
Diagram editing and versions	2–3
Collaboration, roles and shared projects	3
Dashed assumption style in exports (DR2b)	3
Other diagram types	3–4